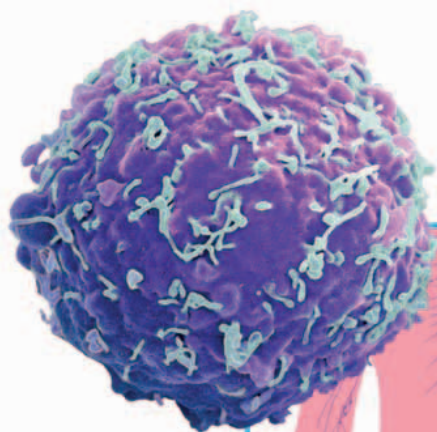


Immune system

Most germs fail to break through the body's barriers, but a few succeed and multiply inside us, causing an infection. When this happens, our immune system fights back. The immune system identifies new germs as enemies and then targets them specifically. It also remembers them for the future, giving us immunity to the diseases they cause.

WOW!

Some sneaky germs continually change their surface markers to escape detection by the immune system.



◀ **ANTIBODY CELL**
Antibodies are made by specialized white blood cells called B-cells.

HOW ANTIBODIES KILL GERMS

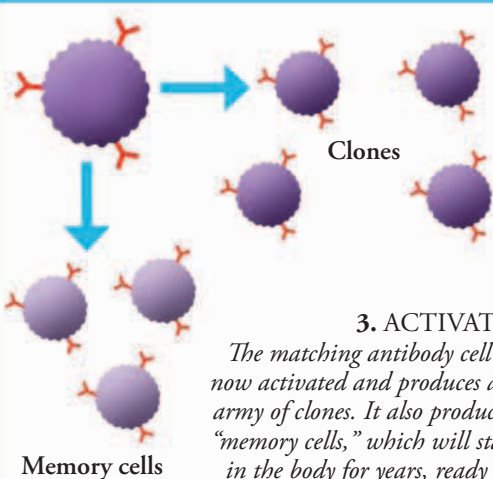
Antibodies are chemicals that stick to germs such as bacteria, flagging them for destruction. There are millions of different types of germs, and each type of antibody recognizes only one of them. Fortunately, the human body can produce ten billion different types of antibodies, ensuring there's an antibody to match every germ you ever encounter.

1. INFECT A germ invades the body and begins to multiply. It gets carried by body fluids to a lymph node.

Germ

Antibody cells

2. DETECT Inside the lymph node, many different antibody cells touch the germ to try and identify the unique molecules on its surface. Eventually, a cell with matching molecules recognizes the germ as an enemy and sticks to it.



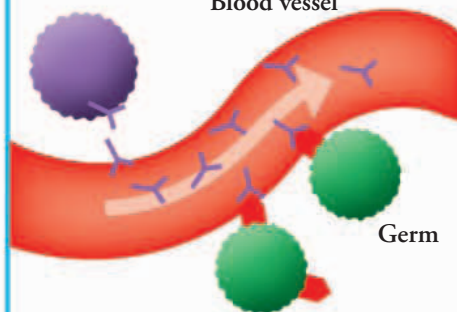
3. ACTIVATE

The matching antibody cell is now activated and produces an army of clones. It also produces "memory cells," which will stay in the body for years, ready to mount a swift attack if the same germ ever comes back.

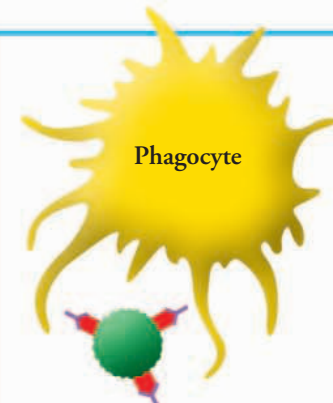
4. SEEK The clones make antibodies to match the unique molecules on the germ's surface. The antibodies are released into the blood and travel throughout the body. Wherever they find matching germs, they stick to them.

Blood vessel

Germ



Phagocyte



5. DESTROY The antibodies act as beacons to white blood cells called phagocytes, which swallow and destroy the germs.